SMART WARDROBE ORGANIZATION SYSTEM

Project ID – R26-IT-053

PROJECT PROPOSAL REPORT

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

The rapid growth of the fashion industry and the increasing number of clothing choices available to individuals have created a common problem known as decision fatigue, where people spend significant time deciding what to wear. While many existing fashion recommendation systems focus on suggesting new items for purchase, they rarely support the efficient management and utilization of a user's existing wardrobe. This creates a gap in personal wardrobe organization and sustainable clothing usage.

This research proposes the development of a **Smart Wardrobe Organization System**, an AI-based solution designed to automatically classify, organize, and analyse clothing items within a digital wardrobe. The system utilizes machine learning models to analyse wardrobe images and identify key attributes such as clothing category, colour, pattern, and fabric. These attributes are then stored in a structured wardrobe database that enables intelligent organization and efficient item management.

A key feature of the proposed system is **Intelligent Wear Count Tracking**, which automatically records how frequently each clothing item is worn. This allows the system to identify frequently used, rarely used, and forgotten wardrobe items. By analysing usage patterns, the system provides insights that help users rotate their clothing more effectively and reduce outfit repetition.

The system also introduces a **Laundry Logic mechanism**, which tracks the cleanliness status of clothing items. After an item is worn, it is marked as dirty and temporarily excluded from outfit suggestions until it is cleaned. This ensures that outfit planning modules only recommend clean and available items.

In addition, the system generates wardrobe analytics such as usage frequency, duplicate clothing detection, and sustainability insights. By highlighting underutilized items and encouraging their reuse, the system helps reduce unnecessary clothing purchases and promotes more sustainable fashion behaviour.

The proposed Smart Wardrobe Organization System contributes to the development of intelligent wardrobe management solutions by integrating machine learning, behavioural analytics, and practical wardrobe tracking mechanisms to create a personalized and efficient clothing organization platform.

Keywords: Smart Wardrobe, Clothing Classification, Machine Learning, Wear Count Tracking, Sustainable Fashion, Wardrobe Management.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
CNN	Convolutional Neural Network
CV	Computer Vision
API	Application Programming Interface
UI	User Interface
NLP	Natural Language Processing
DB	DataBase
JSON	JavaScript Object Notation
SQL	Structured Query Language
IOT	Internet of Things

1. INTRODUCTION

1.1 Background

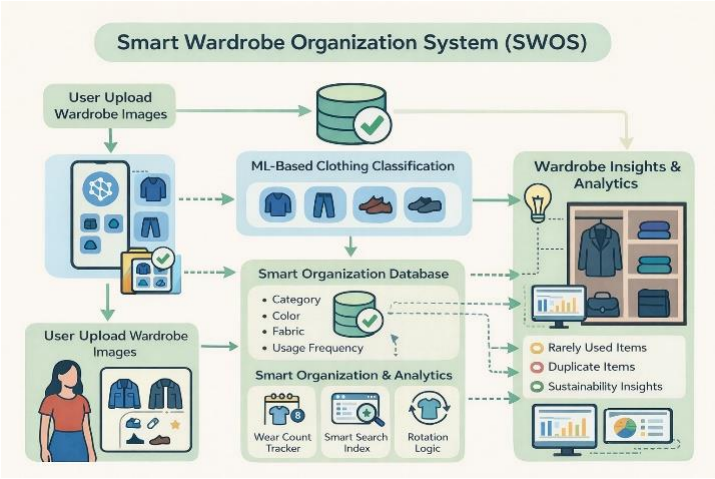
The fashion industry has experienced rapid growth in recent years due to the rise of e-commerce platforms and fast fashion trends. As clothing becomes more accessible and affordable, individuals tend to purchase more garments than ever before. While this increase in clothing availability provides greater fashion choices, it also introduces new challenges in managing personal wardrobes effectively.

Many individuals today own large numbers of clothing items but struggle to organize and utilize them efficiently. As a result, people often spend a considerable amount of time deciding what to wear each day. Studies show that individuals may spend several minutes every morning selecting an outfit, which accumulates to a significant amount of time over a year. This phenomenon is commonly referred to as **decision fatigue**, where users become overwhelmed by the number of available clothing options.

Another common issue is that people tend to repeatedly wear a small subset of their wardrobe while many other items remain unused for long periods. In many cases, individuals forget about certain clothing items simply because they are not easily visible or organized. This leads to inefficient wardrobe usage and contributes to unnecessary clothing purchases.

Traditional wardrobe organization methods are mostly manual and rely on physical arrangements such as sorting clothes by type or color inside a closet. Although these methods provide basic organization, they do not offer intelligent insights into wardrobe usage patterns or clothing compatibility. Furthermore, manual organization does not provide data about how frequently specific clothing items are worn or which items remain unused.

With advancements in **Artificial Intelligence (AI), Machine Learning (ML), and Computer Vision**, digital wardrobe systems have emerged as innovative solutions to these challenges. A digital wardrobe allows users to upload images of their clothing items and store them in an organized database. Machine learning models can analyze these images and automatically classify garments based on attributes such as clothing type, color, pattern, and fabric. This automation reduces manual effort while enabling more structured wardrobe management.



Within the broader E-Wardrobe system, the **Smart Wardrobe Organization System** plays a critical role in managing and structuring clothing inventories. Unlike simple digital closets that only store clothing images, the Smart Wardrobe Organization System introduces intelligent mechanisms for organizing and analyzing wardrobe data. The system automatically groups clothing items into categories and allows users to search, filter, and retrieve items efficiently.

One key innovation in this component is the **Intelligent Wear Count Tracking mechanism**, which records how frequently each clothing item is worn. By analyzing usage frequency and the last worn date, the system can identify clothing items that are rarely used. This allows the system to generate insights about wardrobe usage patterns and encourage users to utilize underused items more frequently.

Another important feature is **Smart Search Indexing**, which enables efficient retrieval of clothing items based on attributes such as color, category, occasion, or wear frequency. Instead of manually browsing through an entire wardrobe collection, users can quickly locate specific items through indexed search queries.

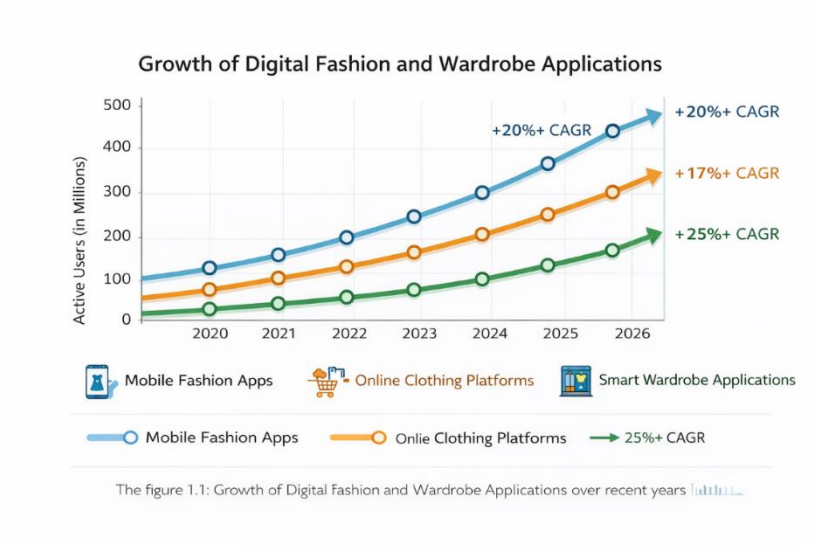
In addition to improving wardrobe organization, the Smart Wardrobe Organization System also supports **sustainable fashion practices**. The global fashion industry is one of the largest contributors to environmental pollution due to excessive clothing production and waste. Many consumers purchase new clothing items simply because they are unaware of the garments already available in their wardrobes. By analyzing clothing usage patterns and identifying rarely worn items, the system encourages users to reuse existing clothing rather than purchasing new items unnecessarily.

Furthermore, the system introduces a **Sustainability and Rotation Algorithm**, which identifies "forgotten" wardrobe items that have not been worn for a long period. These items can then be prioritized in outfit recommendations to promote wardrobe diversity and reduce fashion over-consumption.

The Smart Wardrobe Organization System also integrates with other modules of the E-Wardrobe platform, including the outfit recommendation and visualization components. By maintaining a well-structured wardrobe database, the organization system provides accurate and reliable data that can be used by other modules to generate outfit suggestions and personalized fashion insights.

Overall, the Smart Wardrobe Organization System aims to transform traditional wardrobe management into an intelligent digital solution. By combining machine learning-based clothing

classification, usage tracking, and data analytics, the system enables users to manage their wardrobes more efficiently while promoting sustainable clothing practices.



The figure 1.1: Growth of Digital Fashion and Wardrobe Applications over recent years

The rapid growth of digital fashion technologies has transformed how people manage and select clothing. As shown in Figure 1.1, the number of users of **mobile fashion applications**, **online clothing platforms**, and **smart wardrobe applications** has increased significantly between 2020 and 2026. Mobile fashion apps have grown from around **100 million users to nearly 500 million users**, while online clothing platforms and smart wardrobe applications are also experiencing strong growth.

This trend indicates that consumers are increasingly relying on **digital solutions to organize wardrobes, explore fashion trends, and receive personalized outfit suggestions**. In particular, **smart wardrobe applications** are growing at a rate of more than **25% CAGR**, highlighting the increasing demand for intelligent clothing management systems.

A **Smart Wardrobe Organization System** leverages technologies such as **Artificial Intelligence (AI), Computer Vision, and Machine Learning** to help users digitally manage their clothing items. The system can categorize garments, recommend outfit combinations, and optimize wardrobe organization based on user preferences, seasons, and occasions.

Therefore, the development of a Smart Wardrobe Organization System addresses the modern need for **efficient wardrobe management, personalized fashion recommendations, and enhanced user convenience in digital fashion ecosystems**.

1.1.1 Statistical Evidence

Recent studies and fashion industry reports highlight the importance of intelligent wardrobe management systems in reducing clothing waste and improving clothing usage efficiency.

Research indicates that the average person regularly uses only **20%–30% of the clothes in their wardrobe**, while the remaining **70%–80% remain rarely used or forgotten**. This phenomenon leads to unnecessary clothing purchases and contributes to the growing problem of **fashion over-consumption and textile waste**.

According to sustainability reports in the fashion industry, approximately **92 million tons of textile waste are generated globally each year**, largely due to underutilized clothing items. Smart wardrobe technologies aim to address this issue by encouraging users to maximize the use of existing garments.

In addition, user behavior studies show that **more than 60% of people struggle to decide what to wear**, even when they own many clothing items. This indicates a strong need for **automated wardrobe organization and outfit recommendation systems**.

The proposed **Smart Wardrobe Organization System** integrates **wear-frequency tracking and sustainability rotation algorithms** to solve this problem. By automatically recording how often each item is worn, the system can identify **frequently used and rarely used items**. Statistical analysis of this data allows the system to generate insights such as:

- Percentage of frequently worn items
- Identification of forgotten or rarely used clothing
- Most commonly used clothing categories
- Clothing usage trends over time

Furthermore, the **Laundry Logic integration** ensures that worn clothes are automatically marked as **"dirty"**, preventing them from being recommended again until cleaned. This improves recommendation accuracy and reflects real-life wardrobe management behavior.

Through these data-driven insights, the system promotes **sustainable clothing consumption**, encourages users to **reuse existing items**, and reduces the need for unnecessary clothing purchases.

1.2 Literature Survey

The concept of smart wardrobe systems has gained significant attention in recent years due to advancements in artificial intelligence, computer vision, and machine learning technologies. Researchers have explored several approaches to automate wardrobe management, clothing classification, and outfit recommendation.. Smart wardrobe organization systems integrate digital technologies with clothing management. Previous research has explored RFID-based wardrobe systems that track clothing items using sensors and embedded tags. Although these systems provide automated inventory tracking, they require specialized hardware infrastructure which makes them expensive and difficult to implement.

Recent research focuses on **AI-based wardrobe management applications** that use computer vision to analyze clothing images. These systems extract visual attributes such as color, pattern, and category to organize digital wardrobes.

However, most existing systems lack intelligent usage analytics and sustainability features.

1.2.1 Digital Wardrobe Management Systems

Digital wardrobe management systems allow users to store and organize their clothing items digitally. Early systems required users to manually upload clothing images and categorize them based on type, color, or occasion.

Mobile applications such as **Stylebook and Closet** allow users to maintain a digital wardrobe where clothing items are stored and organized into categories. These systems help users track their wardrobe but rely heavily on manual input and lack intelligent automation.

Because of this limitation, researchers have begun exploring **AI-based wardrobe systems** that automatically detect clothing attributes using computer vision techniques. Digital wardrobe applications allow users to store images of their clothing items and manually categorize them. These applications typically include filters for clothing type or color but rely heavily on manual input.

Recent systems attempt to automate wardrobe organization using AI techniques, but they rarely include features such as **wear frequency tracking, wardrobe analytics, or sustainability recommendations**.

1.2.2 Smart Closet and IoT Wardrobe Systems

Some researchers have proposed **smart closet systems using IoT technologies** such as RFID tags and smart sensors.

These systems track clothing usage and provide wardrobe analytics.

Although these systems provide automation, they have several limitations:

- Expensive hardware requirements
- Complex setup and maintenance
- Limited accessibility for everyday users

Because of these issues, **software-based wardrobe organization systems are becoming more practical research solutions**

1.2.3 Clothing Usage Analytics and Sustainability

Recent research highlights the importance of **tracking clothing usage frequency**.

Many users wear only a small portion of their wardrobe while most clothing items remain unused.

Usage analytics can identify:

• Frequently worn items
• Rarely used clothes
• Wardrobe rotation patterns

These insights help users improve wardrobe utilization and support **sustainable fashion practices** by encouraging re-wearing existing clothes instead of purchasing new items.

1.2.4 Machine Learning for Clothing Classification

Machine learning techniques have been widely used for clothing classification. Convolutional Neural Networks (CNNs) are commonly applied to identify clothing categories, colors, and patterns from images.

Datasets such as **DeepFashion** provide labeled clothing images used for training these models. These classification systems enable automatic tagging of clothing items, which simplifies wardrobe organization.

However, many existing systems focus only on image classification and do not integrate these results into a comprehensive wardrobe management system.

1.2.5 Smart Search and Clothing Retrieval

As digital wardrobes grow larger, efficient search mechanisms become essential.

Smart wardrobe systems use **search indexing techniques** to allow users to retrieve clothing items quickly using attributes such as:

- Color
- Occasion
- Category
- Usage frequency

This approach significantly improves user experience and reduces time spent searching for outfits.

However, many existing systems **do not combine search indexing with intelligent wardrobe analytics.**

1.2.6 Wear Count Tracking and User Behavior Analytics

Tracking how often clothing items are worn is a key innovation in smart wardrobes. Few studies address this area:

- **Usage Analytics in Smart Closets**: Research shows that most users wear only 20-30% of their wardrobe regularly, with the remaining items rarely used.

- **Wear Count Tracking**: Automatically updating wear frequency for each clothing item allows users to see their most and least used outfits.

- **Impact on Fashion Consumption**: Tracking wear counts can influence users to rotate their wardrobe effectively, promoting sustainability by reducing the need for frequent purchases.

Existing digital wardrobe apps often allow manual tagging or logging but **lack automated wear count updates** and fail to suggest underused items intelligently.

1.3 Research Gap

. In recent years, the integration of digital technologies into the fashion and lifestyle industry has led to the development of various wardrobe management applications and smart fashion systems. These systems are designed to assist users in organizing clothing items, planning outfits, and maintaining digital records of their wardrobe collections. With the rapid advancement of technologies such as Artificial Intelligence (AI), Machine Learning (ML), Computer Vision, and Data Analytics, researchers and developers have explored multiple approaches to improve wardrobe management and personalized fashion assistance [2, 22, 23] .

Many existing digital wardrobe applications allow users to upload images of their clothing items and manually categorize them into groups such as shirts, pants, dresses, or accessories. Some systems also provide simple outfit recommendations based on predefined templates or color matching techniques. Research on fashion image recognition and clothing datasets such as DeepFashion has significantly improved clothing detection and retrieval methods [1]. While these systems provide a basic level of wardrobe management support, they still depend heavily on manual data entry and user interaction, which limits their efficiency and scalability, especially for users with large clothing collections.

Another important limitation in current wardrobe management systems is the lack of intelligent wardrobe organization mechanisms. Most existing applications focus primarily on outfit recommendation or fashion shopping assistance rather than effectively organizing and utilizing the clothing items that users already own. Several studies have focused on clothing recognition and product recommendation from images [6, 10, 11]. However, these approaches mainly target online fashion platforms rather than personal wardrobe organization.

Furthermore, many existing systems do not incorporate automatic clothing classification using machine learning techniques. Research in apparel classification and fashion parsing has shown that clothing items can be identified using computer vision and deep learning models [8, 9]. Without automated classification based on attributes such as color, clothing type, fabric type, pattern, or occasion, users must manually organize and tag each item in their wardrobe. This process can be time-consuming and discouraging for users, reducing the practicality and long-term adoption of such systems.

Another critical limitation observed in current wardrobe applications is the absence of intelligent usage tracking mechanisms. Most existing systems do not record how frequently clothing items are

worn or analyze patterns related to clothing usage. Research on fashion compatibility and recommendation systems has focused mainly on predicting suitable outfit combinations using deep learning techniques [3,14,18], but these systems rarely consider real-life wardrobe usage patterns.

Additionally, current wardrobe management technologies rarely provide analytical insights related to wardrobe usage behavior. For example, users are not informed about underutilized clothing items, duplicate clothing categories, or imbalanced wardrobe compositions. The lack of such analytical capabilities prevents users from making informed decisions about wardrobe organization, clothing rotation, or sustainable fashion practices.

Another notable limitation is the absence of smart search and indexing mechanisms for wardrobe retrieval. In many systems, searching for specific clothing items or outfit combinations can be inefficient because items are not indexed based on multiple attributes such as color, usage frequency, category, or occasion. This limits the ability of users to quickly retrieve clothing items that match their specific preferences or situations.

Moreover, modern wardrobe systems rarely address the issue of fashion overconsumption and sustainability. Many people purchase new clothing items even though they already own similar items that remain unused in their wardrobe. Sustainable fashion research highlights the environmental impact of excessive clothing consumption and emphasizes the importance of better wardrobe utilization [4,20, 21].

To address these limitations, there is a significant research gap in the development of Smart Wardrobe Organization Systems that combine automated clothing classification, intelligent organization algorithms, usage tracking, and wardrobe analytics. Such systems should be capable of automatically identifying clothing attributes using machine learning models, organizing wardrobe items dynamically based on multiple features, and maintaining a structured digital wardrobe database.

Furthermore, an intelligent wardrobe organization system should include wear-frequency tracking and sustainability-based rotation algorithms that encourage users to reuse underutilized clothing items. By identifying rarely worn garments and generating compatible outfit combinations for them, the system can help users maximize the usage of their existing wardrobe while reducing unnecessary clothing purchases.

Additionally, integrating smart search indexing mechanisms can significantly improve wardrobe accessibility by allowing users to quickly locate clothing items based on attributes such as color, clothing type, occasion, or wear frequency.

Therefore, the proposed Smart Wardrobe Organization System aims to bridge these research gaps by introducing an automated and intelligent wardrobe management solution that combines machine learning-based clothing classification, dynamic wardrobe organization, intelligent wear tracking, and sustainability-driven insights. This system will provide users with a more efficient, data-driven, and environmentally conscious approach to wardrobe management.

Existing Systems	**Limitations**
Digital wardrobe apps	Require manual organization
Clothing classification systems	Focus only on recognition
IoT smart wardrobes	Expensive hardware
Fashion recommendation systems	Focus on buying new items

1.4 Research Problem

In modern lifestyles, individuals often own a large number of clothing items, making wardrobe management increasingly difficult. Many people struggle to keep their wardrobes properly organized, track clothing usage, and efficiently select outfits for different occasions. As wardrobes grow larger, it becomes challenging for users to remember what clothing items they own, how often they use them, and how those items can be combined effectively with other garments. This lack of proper wardrobe organization often leads to inefficient clothing usage and unnecessary clothing purchases.

Traditional wardrobe management methods rely on **manual organization**, where users physically arrange their clothing items based on categories such as type, color, or season. However, this manual approach does not provide any digital tracking or intelligent insights into clothing usage patterns. Users often forget about rarely used garments that remain hidden in the wardrobe, which results in underutilization of clothing items.

In recent years, several digital wardrobe management applications have been introduced to assist users in storing clothing images and planning outfits. However, most existing systems require users to manually upload and categorize clothing items, which can be time-consuming and inconvenient. These systems also lack **automated classification mechanisms** that can identify clothing attributes such as type, color, pattern, and fabric using machine learning techniques.

Another significant problem in current wardrobe management solutions is the **absence of intelligent wardrobe organization algorithms**. Most systems store clothing items in simple lists or folders without dynamically organizing them based on important attributes such as clothing category, usage frequency, season, or occasion. As a result, users cannot efficiently locate specific clothing items or identify suitable outfit combinations.

Furthermore, existing wardrobe applications rarely include **wear-frequency tracking mechanisms** that record how often clothing items are used. Without tracking this information, users cannot easily identify frequently worn items, rarely used garments, or duplicate clothing categories within their wardrobe. This limitation reduces the ability of users to optimize wardrobe usage and maintain a balanced clothing collection.

Another critical issue is the lack of **analytical insights related to wardrobe utilization**. Most current systems do not provide meaningful feedback such as identifying underused clothing items, suggesting wardrobe rotation strategies, or analyzing clothing usage patterns over time. Without such insights, users may unknowingly continue to purchase new clothing items while many existing garments remain unused.

Additionally, modern fashion consumption patterns contribute to the problem of **fashion overconsumption**, where individuals buy new clothing items frequently even though they already possess similar garments in their wardrobe. The absence of intelligent systems that promote the reuse of existing clothing items contributes to increased textile waste and unsustainable fashion practices.

Another limitation in current wardrobe technologies is the lack of **efficient search and retrieval mechanisms**. Many systems do not implement advanced search indexing methods that allow users to quickly retrieve clothing items based on multiple attributes such as color, category, occasion, or usage frequency. This reduces system usability and makes outfit selection slower and less convenient.

To address these challenges, there is a need for a **Smart Wardrobe Organization System** that can automatically classify clothing items using machine learning models, organize wardrobe items dynamically based on multiple attributes, track clothing usage patterns, and generate intelligent insights about wardrobe utilization.

The proposed Smart Wardrobe Organization System aims to solve these problems by introducing **automated clothing classification, intelligent wardrobe organization, wear-frequency tracking, smart search indexing, and sustainability-focused wardrobe analytics**. By integrating these features, the system will help users efficiently manage their wardrobes, maximize the use of existing clothing items, and make more sustainable fashion decisions.

Gap Area	Current Status	Proposed Solution in This Component
Usage Analytics	Most digital wardrobe apps only store images or manual tags.	Introduce **Intelligent Wear Count Tracking** to monitor clothing usage automatically.
Search and Retrieval	Existing systems provide basic filtering by type or color.	Implement **Smart Search Indexing** allowing fast retrieval based on multiple parameters including color, occasion, fabric, and wear frequency.
Sustainability & Rotation	Rarely integrated; users are unaware of underused items.	Develop **Sustainability and Rotation Algorithm** to identify forgotten items and encourage re-use.
Laundry Management	Clothing status not tracked; dirty clothes may be suggested.	Integrate **Laundry Logic** to exclude dirty items from suggestions until cleaned.
Personalization	Existing solutions are static; recommendations not tailored to individual wardrobe usage.	Provide **usage-based recommendations** to promote wardrobe variety and reduce outfit repetition.

2. OBJECTIVES

2.1 Main Objective

To develop a **Smart Wardrobe Organization System** that automatically organizes clothing items, tracks usage patterns, and provides intelligent insights to improve wardrobe management and sustainability.

2.2 Specific Objectives

• Develop a machine learning-based clothing classification system.
• Organize wardrobe items into categories based on attributes such as color, type, fabric, and usage frequency.
• Maintain a structured digital wardrobe database with real-time updates.
• Track clothing usage frequency using an intelligent wear count tracking mechanism.
• Generate analytics such as rarely used items and frequently worn clothing.
• Implement smart search indexing for efficient wardrobe retrieval.

• Integrate laundry logic to mark worn items as unavailable until cleaned.
• Implement a sustainability-focused clothing rotation algorithm.

3. Methodology

3.1 System Overview

The **Smart Wardrobe Organization System** is a core component of the E-Wardrobe platform that focuses on automatically organizing clothing items and managing wardrobe data intelligently. The system integrates machine learning, structured databases, and analytics modules to transform unorganized wardrobe images into a structured digital wardrobe with useful insights.

The system enables users to upload images of their clothing items, which are then processed using machine learning models to extract attributes such as clothing category, color, pattern, and fabric. These attributes are stored in a centralized wardrobe database where items are automatically organized into meaningful categories.

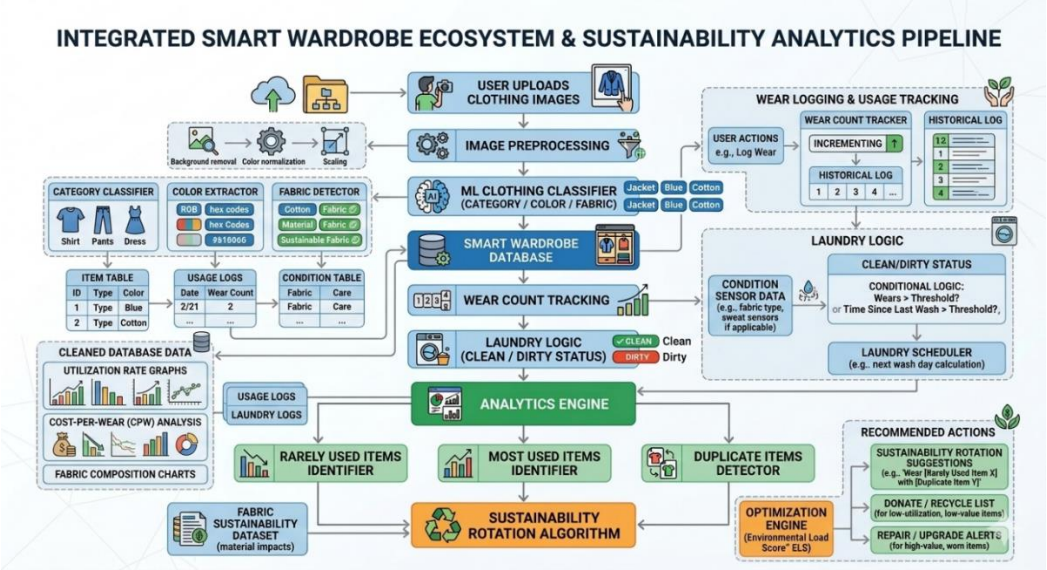
In addition to classification and organization, the system tracks clothing usage through an **Intelligent Wear Count Tracking mechanism**. Each time an outfit is worn, the system records the usage frequency and updates the item's last used date. This data enables the system to identify frequently worn items and rarely used garments.

Another key feature of the system is the **Laundry Logic Integration**, which automatically marks clothing items as “Dirty” after being worn. These items are excluded from outfit recommendations until the user marks them as cleaned, ensuring that outfit suggestions reflect the real availability of clothing.

The system also includes a **Smart Search Indexing mechanism**, allowing users to quickly retrieve wardrobe items based on attributes such as color, clothing type, occasion, or wear frequency. This improves usability and enables efficient wardrobe navigation.

Furthermore, the Smart Wardrobe Organization System includes an **Analytics and Sustainability Engine** that analyzes wardrobe usage patterns. The engine identifies underused clothing items and generates insights such as most worn items, rarely used garments, and clothing category distribution. Based on these insights, a **Sustainability Rotation Algorithm** encourages users to reuse underutilized clothing items, helping reduce fashion over-consumption.

Overall, the Smart Wardrobe Organization System acts as the **data backbone of the E-Wardrobe platform**, providing structured wardrobe data that supports other modules such as outfit recommendation and outfit visualization.



3.2 Wardrobe Image Collection

The first step of the methodology involves collecting wardrobe images from users. Users upload photos of their clothing items through the system interface. These images represent different clothing types such as shirts, trousers, dresses, jackets, and accessories.

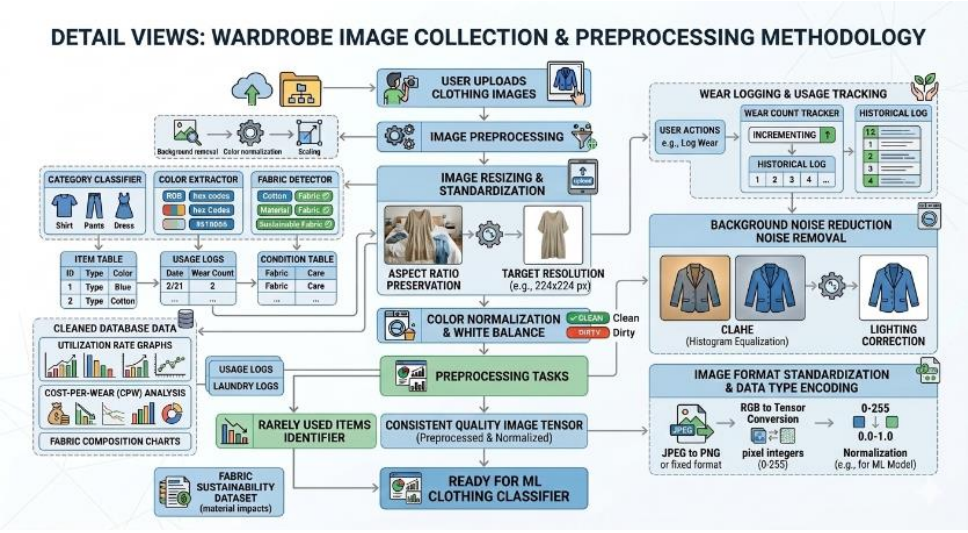
Each uploaded image undergoes preprocessing to ensure consistent quality for machine learning classification.

Preprocessing Tasks

- Image resizing
- Background noise reduction
- Color normalization
- Image format standardization

The system begins by collecting raw images of shirts, trousers, dresses, and accessories uploaded by users via the mobile interface. To ensure these photos are ready for machine learning, they undergo a rigorous **preprocessing pipeline**. First, **format standardization and resizing** transform varied resolutions into a uniform pixel grid (e.g., 224x224). Next, **background noise reduction** uses object segmentation to isolate the garment from household clutter. Finally, **color normalization** corrects for inconsistent lighting, while **feature encoding** converts the visual data into numerical tensors. This process ensures the

ML Classifier receives high-signal, standardized data for accurate category and fabric detection.



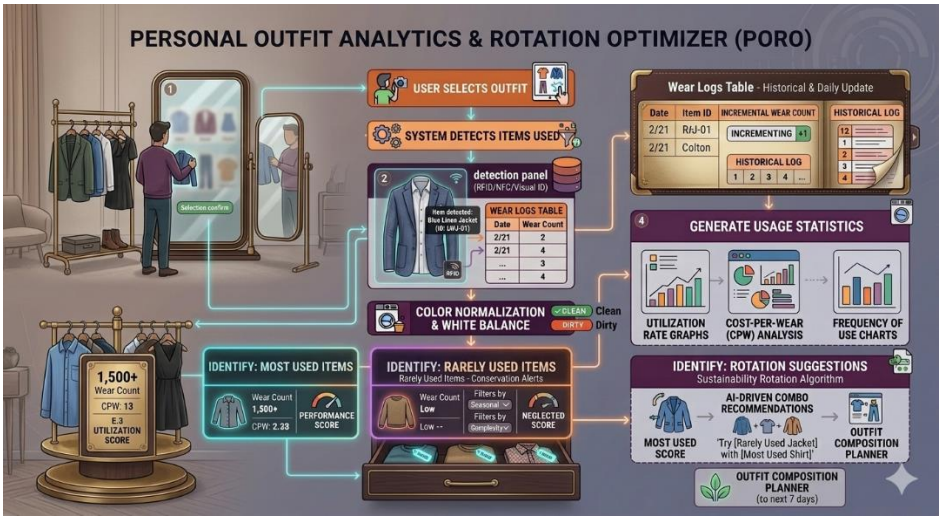
3.3 Intelligent Wear Count Tracking

A key innovation in this component is the **Intelligent Wear Count Tracking System**. Every time a user selects an outfit, the system automatically updates the wear count of each clothing item used in the outfit.

The system stores the following information:

- Wear frequency
- Last worn date
- Outfit combinations used
- Usage statistics

These metrics help identify frequently used clothing items as well as rarely used items.



3.4 Smart Search Indexing

Searching for specific clothing items within large wardrobes can be time-consuming. To address this issue, the system implements a **Search Indexing Mechanism** that allows users to quickly find clothing items based on multiple parameters.

Search filters include:

• Clothing category
• Color
• Fabric type
• Occasion
• Wear frequency

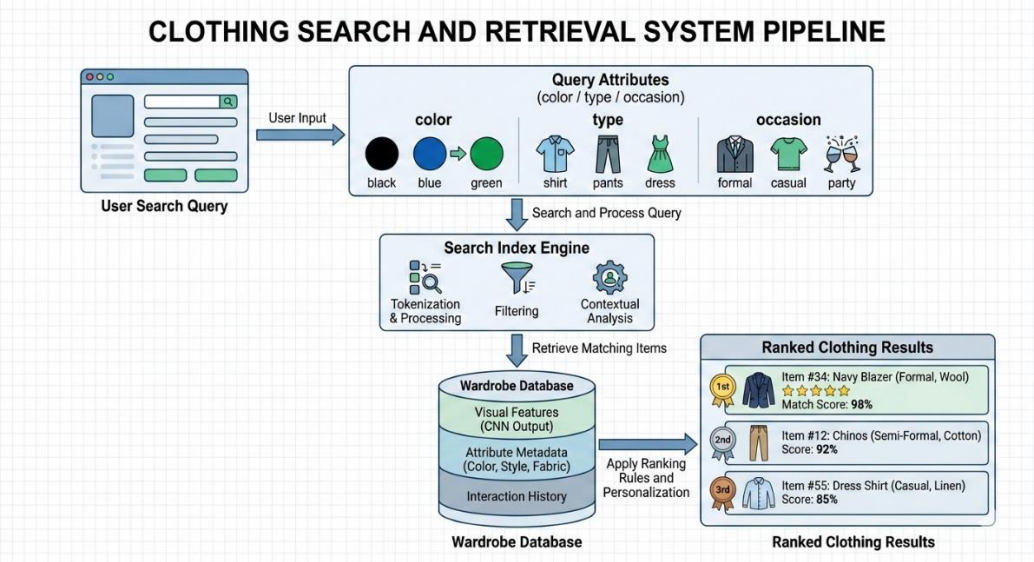
This indexing mechanism improves usability and allows users to retrieve relevant clothing items instantly.

 User Search Query (Input): The process begins with the user providing specific parameters such as **Color** (e.g., green), **Type** (e.g., shirt), and **Occasion** (e.g., formal).

 Search Index Engine (Processing): This acts as the "brain." It breaks down the query through **tokenization**, filters the data, and performs **contextual analysis** to understand the user's intent beyond just keywords.

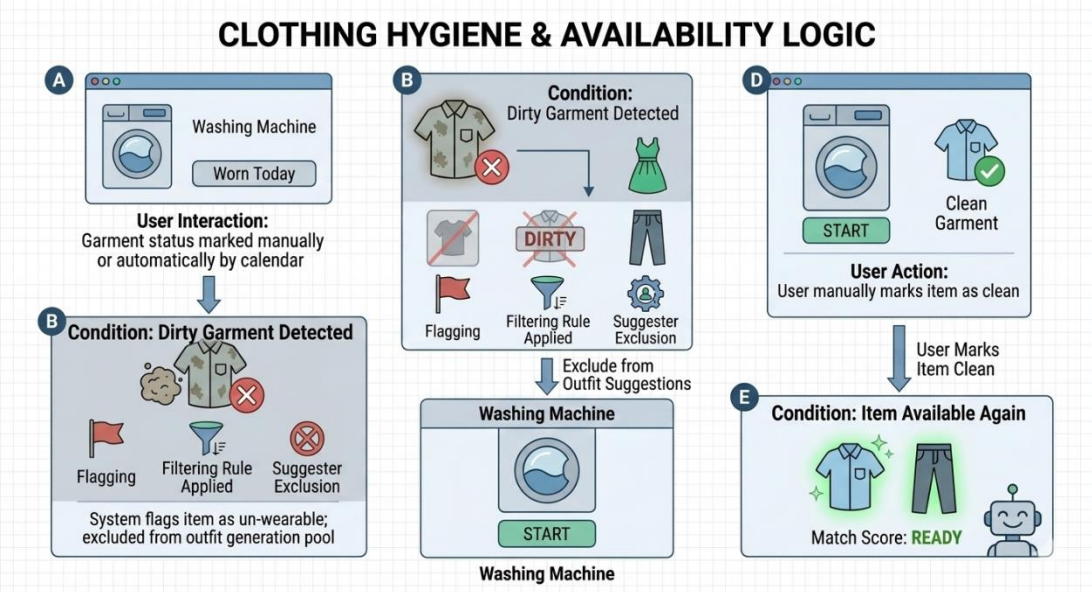
 Wardrobe Database (Storage & Retrieval): The system queries a specialized database. This database stores more than just text; it contains **Visual Features** (analyzed by AI/CNN), **Attribute Metadata** (fabric, style), and the user's **Interaction History**.

 Ranked Clothing Results (Output): Finally, the system applies ranking rules and personalization. Instead of a random list, it returns a curated selection with **Match Scores**, prioritizing the items that best fit the search criteria and the user's style.



3.5 Laundry Logic Integration

The system integrates a **Laundry Status Feature** to ensure that dirty clothing items are not recommended.and The system excludes dirty items from outfit suggestions until the user marks them as cleaned. To maintain realistic wardrobe management, the system includes **Laundry Logic**. After an item is worn, it is automatically marked as **Dirty**. Dirty items cannot appear in outfit recommendations until they are cleaned.

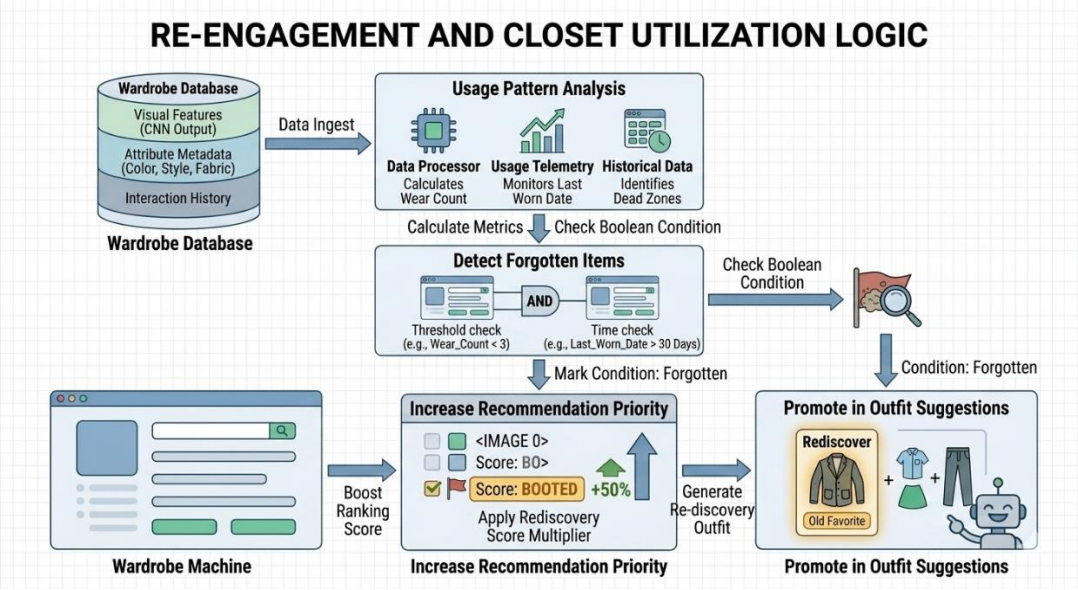


- **A. Status Trigger (Input):** The process begins when a garment's status changes. This can happen **manually** (user clicks "Worn Today") or **automatically** (integration with a digital calendar or smart laundry basket sensors).

- **B. Automated Flagging & Filtering:** Once an item is detected as "Dirty," the system applies a **filtering rule**. It attaches a metadata flag to the item in the database, effectively "hiding" it from the recommendation engine.
- **C. Outfit Exclusion:** The AI Suggester excludes these flagged items from its "Outfit Generation Pool." This ensures the user is never recommended a dirty shirt for a clean outfit, maintaining the system's reliability and hygiene logic.
- **D. The "Clean" Reset:** When the user completes a laundry cycle and marks the item as "Clean" in the app, the system performs a **status update**.
- **E. Availability Restoration:** The item is re-indexed into the searchable database. Its "Match Score" is reactivated, making it instantly available for future outfit suggestions and search results.

3.6 Sustainability and Rotation Algorithm

To address fashion over-consumption, the system implements a Wardrobe Rotation Algorithm. This algorithm identifies clothing items that have not been worn for long periods and promotes them in outfit suggestions.



- **Wardrobe Database (Data Source):** The cycle begins by ingesting raw data from the central database, which contains **Visual Features** (extracted via AI), **Metadata** (fabric/style), and **Interaction History** (when the item was last touched).
- **2. Usage Pattern Analysis (The Processor):** The system runs a background audit on all inventory. It calculates specific **telemetry metrics** for each item, primarily focusing on how many times an item has been worn and how much time has passed since its last appearance.
- **3. Detect Forgotten Items (The Logic Gate):** This stage applies a strict boolean condition. If an item meets both criteria—low usage frequency (**Wear_Count < Threshold**) and long-

term inactivity (**Last_Worn_Date > 30 Days**) — it is officially flagged as a "Forgotten Item."

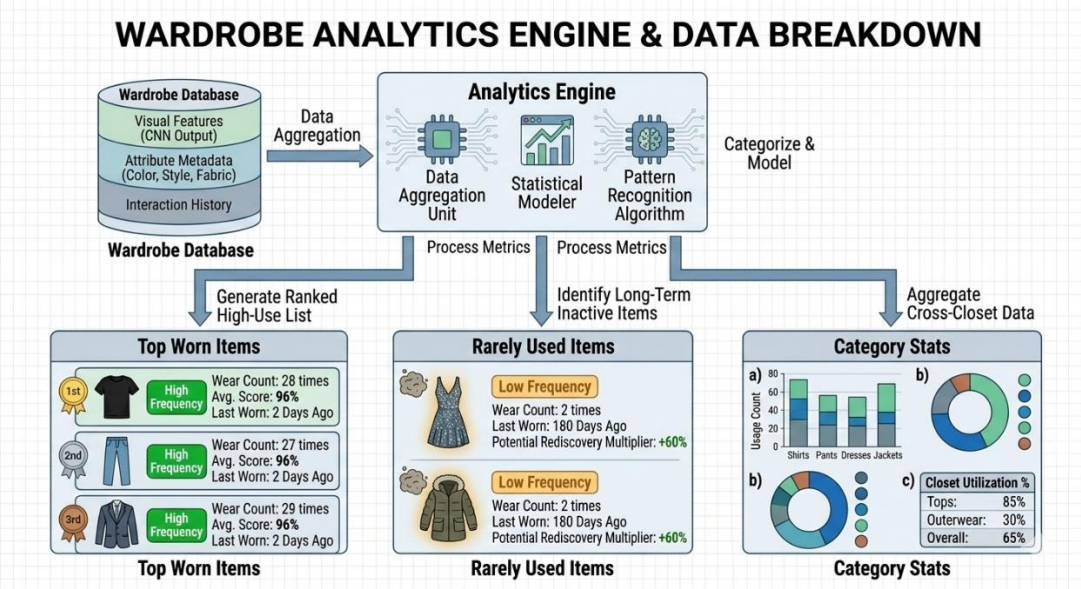
- **4. Increase Recommendation Priority (The Optimizer):** To rescue the item from the back of the closet, the system applies a **Score Multiplier** (e.g., +50% boost). This heuristic adjustment ensures the item ranks higher than newer or more frequently worn clothes during the next suggestion cycle.
- **5. Promote in Outfit Suggestions (The Output):** The final step sees the "Forgotten Item" surfaced to the user as a **"Rediscover" recommendation**. The AI styler builds a fresh outfit around this specific piece, successfully re-integrating it into the user's active rotation.

3.7 Wardrobe Analytics Engine

The system continuously analyzes wardrobe data to generate insights about clothing usage.

Analytics Generated:

• Most frequently worn clothing
• Rarely used clothing
• Clothing category distribution
• Duplicate clothing detection



The **Wardrobe Analytics Engine** acts as the central intelligence layer of the smart organization system, transforming raw inventory data into actionable user insights through a structured three-tier processing model. The cycle begins with **Data Aggregation**, where the system ingests comprehensive information from the Wardrobe Database, including visual features, metadata, and historical interaction logs. This data is then passed through a specialized **Analytics Engine** consisting of a Data Aggregation Unit, a Statistical Modeler, and a Pattern Recognition Algorithm. This engine categorizes and models the information to identify specific clothing behaviors and closet health metrics.

Once the data is processed, the system generates three distinct outputs for the user's dashboard. The first output identifies **Top Worn Items**, creating a ranked high-use list that highlights the user's "essentials" based on high wear counts and frequency scores. Simultaneously, the engine detects **Rarely Used Items**, pinpointing garments with low-frequency telemetry and long-term inactivity, which are then flagged for the rediscovery multiplier mentioned in earlier components. Finally, the system aggregates cross-closet data to produce **Category Stats**, providing a macro-view of the wardrobe through visualization tools like closet utilization percentages and category-specific usage charts. This holistic process ensures the user understands not only what they own but also how their wearing habits impact the overall efficiency of their wardrobe.

3.8 Evaluation Metrics

Classification accuracy

measures the proportion of correctly classified clothing items compared to the total number of items processed by the system.

Accuracy = Correct Predictions / Total Predictions

Purpose

- Evaluates how accurately the system identifies clothing attributes such as category, color, and pattern.

Wardrobe Organization Efficiency

Measures how effectively the system organizes clothing items in the digital wardrobe.

Metric Example

- Time taken to categorize clothing
- Percentage of items automatically categorized

Wear Count Tracking Accuracy

Evaluates whether the system correctly records how often an item is worn.

Metric

- Correct wear count updates
- Correct laundry status updates

System Response Time

System response time measures how quickly the system processes clothing images and retrieves wardrobe items.

Purpose

- Evaluates system performance and user experience.
- Ensures the wardrobe system operates efficiently.

F1 Score

The F1 Score is the harmonic mean of precision and recall and provides a balanced measure of classification performance.

F1 Score = 2 × (Precision × Recall) / (Precision + Recall)

Purpose

- Provides a balanced evaluation when dealing with classification tasks.
- Useful when the dataset contains imbalanced clothing categories.

Precision

Precision measures the proportion of correctly predicted clothing items among all items predicted for a specific category.

Precision = True Positives / (True Positives + False Positives)

Purpose

- Evaluates the correctness of the system when predicting clothing categories.
- Reduces incorrect classification results.

Recall

Recall measures how many relevant clothing items are correctly detected by the classification system.

Recall = True Positives / (True Positives + False Negatives)

Purpose

- Determines whether the system successfully identifies all clothing items belonging to a particular category

4.PROJECT REQUIREMENTS

4.1 Functional Requirements

1. Clothing Image Upload

This function allows users to upload images of their clothing items into the digital wardrobe system. The uploaded images are stored in the database and used by the system for further processing such as classification and organization.

2. Automatic Clothing Classification

The system uses machine learning models to automatically identify clothing attributes such as category (shirt, pants, dress), color, pattern, and fabric type. This helps the system understand the clothing item without requiring manual input from the user.

3. Smart Wardrobe Organization

After classification, the system organizes clothing items into structured digital categories. This helps users manage their wardrobe efficiently and quickly locate specific clothing items.

4. Wear Count Tracking

The system tracks how often each clothing item is worn by the user. Each time an item is selected or used in an outfit, the wear count is updated in the database to monitor clothing usage patterns.

5. Laundry Status Management

When a clothing item is worn, the system marks it as **“dirty.”** These items are temporarily excluded from outfit suggestions until the user updates the status to **“clean”** after washing.

6. Smart Search Indexing

The system provides a fast search feature that allows users to find clothing items using filters such as color, category, occasion, or wear frequency. This improves the efficiency of wardrobe browsing.

7. Usage Analytics Generation

The system analyzes wardrobe data and generates useful insights such as the most frequently worn items, rarely used clothes, and overall wardrobe usage statistics. These analytics help users better understand their clothing habits.

8. Wardrobe Update Management

The system allows users to manage their wardrobe by adding new clothing items, updating item information, or deleting items that are no longer in use. This ensures the digital wardrobe remains accurate and up to date.

4.2 Non Functional Requirements

1.Performance

Performance refers to how fast the system processes user actions. The Smart Wardrobe Organization System should quickly classify clothing images and organize them into categories within a few seconds after the user uploads an item.

2.Scalability

refers to the system's ability to handle an increasing number of wardrobe items. The system should continue to work efficiently even when users upload a large number of clothing items into their digital wardrobe.

3. Usability

Usability focuses on how easy the system is for users to interact with. The interface should be simple, clear, and easy to navigate so that users can upload clothing items, search the wardrobe, and manage their clothing without difficulty.

4. Reliability

Reliability ensures that the system functions correctly and consistently over time. The system should accurately store clothing information and prevent data loss so that users can trust the wardrobe data.

5. Security

Security refers to protecting user data and wardrobe information from unauthorized access. The system should securely store clothing images and personal wardrobe data using proper authentication and data protection methods.

6. Availability

Availability means that the system should be accessible whenever users need it. Users should be able to log in and access their digital wardrobe at any time without major system downtime.

4.3 User Requirement

1. **Digital Wardrobe Organization**
   Users should be able to digitally organize their clothing items into structured categories within the system.

2. **Easy Clothing Image Upload**
   Users should be able to easily upload images of their clothing items to the digital wardrobe.

3. **Quick Search and Filtering**
   Users should be able to quickly search and filter clothing items based on attributes such as color, category, occasion, or usage frequency.

4. **Wardrobe Usage Analytics**
   Users should be able to view wardrobe analytics such as most used clothes, least used clothes, and clothing usage patterns.

5. **Rarely Used Item Alerts**
   Users should receive notifications or alerts for rarely worn clothing items to encourage better wardrobe utilization.

6. **Laundry Status Management**
   Users should be able to mark clothing items as clean after laundry so they can be reused in outfit suggestions**.**

7. **Wear Frequency Tracking**
   Users should be able to see how often each clothing item has been worn through the wear count tracking feature**.**

8. **Wardrobe Item Management**
   Users should be able to add, edit, or remove clothing items from their digital wardrobe.

9. **Personalized Wardrobe Insights**
   Users should receive insights about wardrobe usage trends and recommendations to improve clothing rotation.

10. **User-Friendly Interface**
    Users should be able to interact with the system through a simple and easy-to-use interface**.**

4.4 System Requirements

Hardware Requirements:

1. **1. Computer or Mobile Device**

A computer, laptop, or mobile device is required for users to access the system interface and manage their digital wardrobe. Developers also need a computer system to build, train, and test the machine learning models used for clothing classification.

2. **2. Minimum 8GB RAM**

At least **8GB of RAM** is recommended for running machine learning models and processing clothing images. Adequate memory ensures smooth execution of image classification algorithms and prevents system performance issues during model training and testing.

3. **3. Storage for Clothing Image Dataset**

Sufficient storage space is required to store clothing images uploaded by users and datasets used for training the machine learning models. This storage may be local storage or cloud-based storage depending on system architecture

4. 4. Internet Connection

A stable internet connection is required for accessing cloud services, updating the system, storing images remotely, and enabling communication between the frontend interface and backend services.

5. 5. GPU (Optional but Recommended)

A Graphics Processing Unit (GPU) can significantly improve the performance of machine learning model training and image processing tasks. GPUs accelerate deep learning operations and reduce training time.

Software Requirements

1. Programming Language – Python

Python is used as the main programming language for developing the system. It is used to build the machine learning model, process clothing images, and handle backend logic.

2. Machine Learning Framework – TensorFlow / PyTorch

TensorFlow or PyTorch is used to develop and train the machine learning model that can recognize and classify different types of clothing items from uploaded images.

3. Backend Framework – Flask / Django

Flask or Django is used to build the backend of the system. It handles user requests, processes image uploads, manages wardrobe data, and connects the machine learning model with the database.

4. Database Management System – MySQL / MongoDB

MySQL or MongoDB is used to store wardrobe information such as clothing images, item categories, usage history, and user preferences.

5. Frontend Technologies – React / HTML / CSS / JavaScript

These technologies are used to create the user interface of the system. They allow users to upload clothing images, view wardrobe items, search for clothes, and access wardrobe analytics.

6. Image Processing Library – OpenCV

OpenCV is used for image processing tasks such as resizing images, improving image quality, and preparing clothing images for machine learning classification.

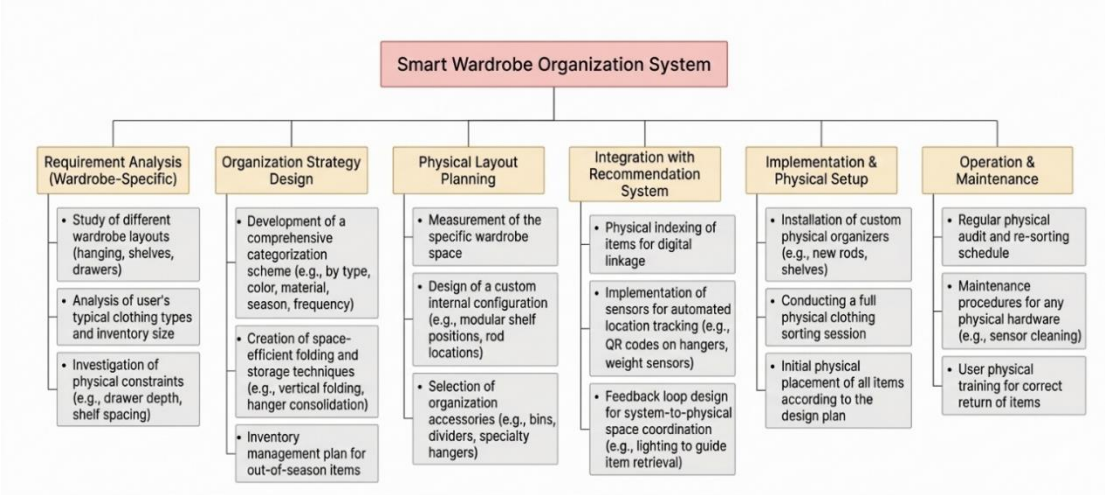
7. API Communication – RESTful API

REST APIs are used to enable communication between the frontend interface and the backend system for data exchange and system functionality.

4.5 EXPECTED TEST CASES

TestCase ID	Test Description	Expected Result
TC01	Upload clothing image	Image successfully uploaded to wardrobe
TC02	Automatic classification of clothing item	System correctly identifies clothing category
TC03	Smart wardrobe organization	Item stored in correct wardrobe category
TC04	Wear count update	Wear frequency increases when item is used
TC05	Laundry status update	Item marked as dirty after use
TC06	Search functionality	Correct clothing items retrieved based on filters
TC07	Rarely used item detection	System identifies items with low usage frequency
TC08	Wardrobe analytics generation	System generates usage statistics

4.6 Work Breakdown Structure



4.7 Gant Chart

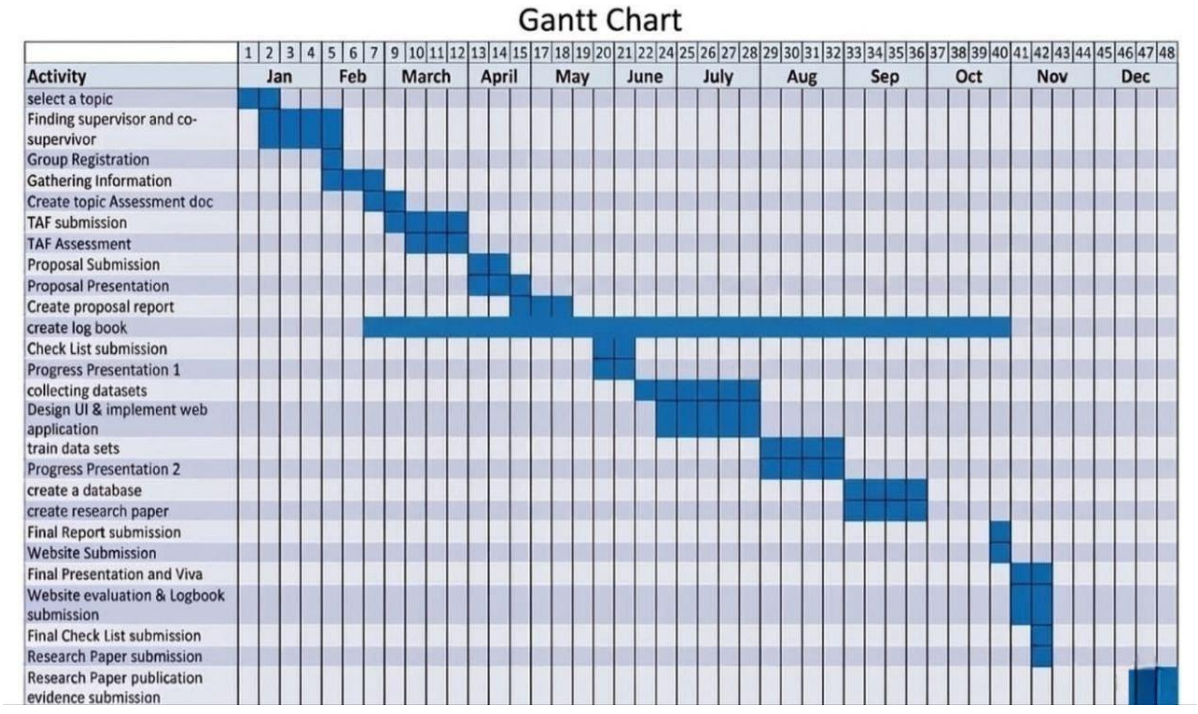


Figure 4. 2 Project Timeline Gantt Chart

4.8 Budget Estimation

The development of the Smart Wardrobe Organization System requires several software tools, cloud services, and hardware resources. These resources are necessary for developing the mobile application, implementing machine learning models, performing image processing tasks, and storing project data. The software costs include development environments, machine learning frameworks, AR development tools, and supporting libraries required for mobile application development, deep learning model implementation, and image processing. Although some tools such as Python, Blender, and Unity provide free versions, additional plugins, development utilities, and assets may require licensing or subscription payments. Cloud services are required for model storage, data hosting, and system deployment. Internet connectivity is also essential for downloading datasets, accessing cloud services, and collaborating with development platforms.

Estimated amount for a 6-month period (SL rupees)

Item	
Licensed Software Tools	10,000.00
Cloud Services / Hosting	10,000.00
Online Courses / Workshops	10,000.00
External Storage	10,000.00
Internet / Networking	12,000.00
Other Hardware	10,000.00
Total	**62,000.00**

5.DESCRIPTION OF PERSONNEL AND FACILITIES

5.1 Personnel

This component of the research is carried out independently by Mayadunna R.H.A.M.A**,** Student ID – IT22067142**, an undergraduate student pursuing the** BSc. (Hons) in Information Technology specializing in Information Technology at the Sri Lanka Institute of Information Technology (SLIIT).

The researcher is responsible for the design, development, implementation, and evaluation of the **Smart Wardrobe Organization System**, which is a key component of the **AI-Driven Outfit Recommendation and Matchmaking System**. This component focuses on organizing and managing digital wardrobe items, enabling intelligent clothing categorization, tracking usage patterns, and assisting the recommendation engine through structured wardrobe data.

The researcher independently handles all stages of development including **system design, machine learning model implementation, database architecture, and evaluation of the wardrobe management algorithms**.

Researcher Responsibilities:

Smart Wardrobe Data Management

The researcher designs and develops the digital wardrobe management framework that allows users to store and manage their clothing items efficiently. Each wardrobe item is digitally represented using metadata such as clothing category, color, material, brand, season, and usage frequency.

The system supports automatic organization and indexing of wardrobe items, enabling efficient retrieval and filtering of clothing data.

Clothing Image Classification

The researcher develops a computer vision pipeline to automatically identify and classify clothing items from uploaded images.

This process includes:

- Image preprocessing
- Feature extraction

- Clothing category classification
- Color and texture detection

Deep learning models such as Convolutional Neural Networks (CNNs) are utilized to improve the accuracy of clothing recognition.

Wardrobe Organization Algorithm Development

The researcher designs algorithms that automatically organize clothing items within the smart wardrobe. These algorithms categorize clothing based on attributes such as:

- Clothing type (shirts, pants, jackets, dresses)
- Color palette
- Seasonal suitability
- Formality level
- Style category

The organization system enables users to quickly locate items and supports efficient wardrobe navigation.

Wear Frequency Tracking

The researcher implements a wear frequency monitoring module that tracks how often each clothing item is used.

The system analyzes usage patterns and provides insights such as:

- Frequently worn items
- Rarely used garments
- Wardrobe utilization statistics

These insights help users better manage their wardrobe and avoid unnecessary clothing purchases.

Laundry and Sustainability Tracking

To promote sustainable fashion practices, the researcher develops a laundry tracking mechanism that monitors clothing usage and recommends appropriate washing intervals.

The system helps reduce unnecessary washing cycles and encourages responsible garment care, thereby extending clothing lifespan and reducing environmental impact.

Database Design and Data Integration

The researcher designs and implements the wardrobe database architecture using modern data management technologies.

The system uses:

- MongoDB for storing clothing metadata and wardrobe information.
- Neo4j Graph Database for representing relationships between clothing items and outfit combinations.

Graph-based relationships enable the system to model compatibility between garments, allowing future integration with recommendation algorithms.

Search and Indexing Mechanisms

The researcher develops efficient search and indexing mechanisms to improve wardrobe accessibility.

The system supports multiple search filters including:

- Clothing category
- Color
- Season
- Occasion
- Usage frequency

These features improve the usability of the Smart Wardrobe System**.**

System Integration

The researcher integrates the Smart Wardrobe component with the broader AI-Driven Outfit Recommendation System.

This includes:

- Developing REST APIs for communication between modules
- Enabling wardrobe data sharing with the recommendation engine
- Supporting user interface interactions for wardrobe browsing and management

System Evaluation and Testing

The researcher conducts comprehensive system testing to evaluate the performance and usability of the Smart Wardrobe System.

Evaluation activities include:

- Functional testing of wardrobe management features
- Accuracy testing of clothing classification models
- User testing to evaluate usability
- Performance testing for database queries and image processing modules

Based on evaluation results, the researcher iteratively improves system functionality and performance.

Supervision

The research project is conducted under the supervision of an academic supervisor from the Department of Information Technology at SLIIT. The supervisor provides academic guidance on:

- Research methodology
- System architecture design
- Machine learning implementation
- Evaluation methodology
- Research documentation

The supervisor ensures that the project meets academic research standards and contributes to ongoing advancements in Artificial Intelligence and Smart Fashion Technologies.

5.2 Facilities

The development of the **Smart Wardrobe Organization System** requires both hardware infrastructure and software tools capable of supporting machine learning development, image processing, and database management

Computing Resources

• **Personal Laptop** equipped with **Intel Core i7 processor and 16GB RAM** used for system development, algorithm implementation, and application testing.

• **AMD Ryzen 5000 Series processor with AMD Radeon Graphics GPU** utilized for running image processing tasks and supporting machine learning model development.

• **Access to SLIIT computing laboratories**, which provide high-performance workstations capable of supporting large dataset processing and experimental evaluation.

• **Cloud computing platforms**, including **Google Colab Pro**, used for training deep learning models with GPU acceleration.

• **MongoDB Cloud Services** used for hosting the wardrobe database and supporting scalable data storage.

Software and Development Tools

Development Environment

• **Visual Studio Code** used for writing and debugging application code.

• **PyCharm Professional** used for advanced Python development and machine learning experimentation.

Machine Learning Frameworks

• **TensorFlow 2.13** used for building and training deep learning models for clothing classification.

• **PyTorch 2.0** used for implementing neural network architectures and experimenting with image recognition models.

Computer Vision Libraries

• **OpenCV** used for image preprocessing, feature extraction, and visual analysis of clothing images.

Database Systems

• **MongoDB** used to store wardrobe item data, clothing attributes, and user preferences.

• **Neo4j Graph Database** used to represent relationships between clothing items and outfit combinations.

Version Control and Collaboration

• **Git** used for version control and source code management.

• **GitHub** used for remote repository hosting and collaborative development.

Project Management Tools

• **Trello** used for organizing development tasks and tracking project progress.

• **Notion** used for project documentation, research notes, and planning.

Documentation Tools

• **Microsoft Office Suite** used for writing research documentation.

• **LaTeX** used for preparing technical research reports and formatting academic documents.

Dataset Access

The system development uses several datasets for training and evaluating machine learning models.

• **DeepFashion Dataset** – Used for clothing image classification and feature extraction.

• **Polyvore Dataset** – Used for studying outfit compatibility and fashion recommendation patterns.

• **Custom Wardrobe Dataset** – A dataset collected through user wardrobe images to simulate real-world clothing collections.

Internet and Connectivity

• **High-speed internet connection** used for accessing cloud computing services, downloading datasets, and conducting model training.

• Access to academic research databases such as **IEEE Xplore** and **ACM Digital Library** through the **SLIIT library**, enabling the researcher to review relevant literature and support the research study.

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